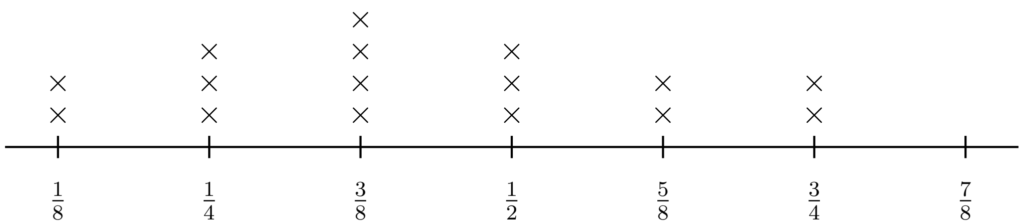
Reading a Line Plot with Fractions

Reading the plot, combining fractions, spread and centre, and redistributing amounts

Directions

- Every problem reads from the Beaker Plot below. Each × stands for one beaker.
- Write fractions in lowest terms, and write amounts greater than one as a mixed number.
- Show the addition or subtraction you did before writing the answer.



Beaker Plot — litres of water in each beaker in the science cupboard.
One × is one beaker. Nothing is recorded at seven eighths.

1. How many beakers hold exactly $\frac{3}{8}$ of a litre?

Answer: __________________

2. How many beakers are recorded on the plot altogether?

Answer: __________________

3. Which amount was recorded more often than any other?

Answer: __________________

4. The two beakers at $\frac{5}{8}$ of a litre are emptied into one jug. How much water is in the jug?

Answer: ________________ L

5. What is the difference between the greatest amount and the least amount on the plot?

Answer: ________________ L

6. How much water is held altogether by the beakers whose amounts are less than $\frac{1}{2}$ of a litre?

Answer: ________________ L

7. The four beakers at $\frac{3}{8}$ of a litre are emptied into one jug and then shared equally between 3 empty beakers. How much water is in each of those 3 beakers?

Answer: ________________ L

8. What is the median of the 16 recorded amounts?

Answer: ________________ L

9. How much water is in the science cupboard altogether?

Answer: ________________ L

10. How much more water is held by all the beakers at $\frac{1}{2}$ of a litre together than by all the beakers at $\frac{1}{4}$ of a litre together?

Answer: ________________ L

11. How many beakers hold more than $\frac{1}{2}$ of a litre?

Answer: ________________

12. One more beaker, holding $\frac{7}{8}$ of a litre, is found and added to the plot.

(a) What is the median of the 17 amounts now?

Answer: ________________ L

(b) Explain why adding the largest amount on the plot did not move the median very far.
